FINITENESS OF H^Q OF THE OUTER TERMS OF A SHORT EXACT SEQUENCE OF LAMBDA-MODULES ON A SEPARATED FINITE TYPE CURVE OVER AN ALGEBRAICALLY CLOSED FIELD IMPLIES FINITENESS OF H^Q OF THE MIDDLE TERM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is the author's complete argument: the long exact sequence application is the whole mechanism, and the claim follows directly from finiteness of the two outer terms over the Noetherian ring.

Editorial note. The label lemma-ses-statements-coefficients defines the statement; the counted proof is the proof environment immediately following the lemma in the source.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217). The class block and the externaldocument lines of the native preamble are not included, because the assembler supplies the wrapper class and those external documents are not part of this material.

Allowed standard prerequisites (editorial).

- The long exact sequence of etale cohomology. Same chapter, the long exact cohomology sequence attached to a short exact sequence 0 -> F_1 -> F -> F_2 -> 0 of sheaves on X_etale. Supplies the exact sequence in which H^q(X, F) sits between H^q(X, F_1) and H^q(X, F_2), which is the entire mechanism of the proof.

Lemma 0.1. *Let Λ be a Noetherian ring, let k be an algebraically closed field, let X be a separated finite type scheme over k of dimension ≤ 1, and let $0 \to \mathcal{F}_1 \to \mathcal{F} \to \mathcal{F}_2 \to 0$ be a short exact sequence of sheaves of Λ-modules on $X_{étale}$. If $H^q_{étale}(X, \mathcal{F}_i)$, $i = 1, 2$ are finite Λ-modules then $H^q_{étale}(X, \mathcal{F})$ is a finite Λ-module.*

Proof. Immediate from the long exact sequence of cohomology. □

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